

AUDIT_IMPLEMENTATION_TRACKER

HelixCode Audit Implementation Tracker

Purpose: Track progress on fixing all issues identified in the Comprehensive Project Audit 2026 **Created:** 2026-01-10 **Status:** Active

Quick Navigation

- [Critical Issues](#)
 - [Test Coverage Tasks](#)
 - [API Implementation Tasks](#)
 - [Feature Completion Tasks](#)
 - [Documentation Tasks](#)
-

Critical Issues (12 Items)

C1: Failing Test - TestExtractPythonSymbols

- **Status:** [] Not Started
- **File:** internal/repomap/repomap_test.go
- **Issue:** TempDir RemoveAll cleanup failure
- **Owner:** *Unassigned*
- **Verified By:** *Pending*

C2: Stub Compression Methods (11 methods)

- **Status:** [] Not Started
- **File:** internal/cognee/performance_optimizer.go:1187-1256
- **Methods:**
 - ☐ NeuralSymbolicCompression.Compress()
 - ☐ NeuralSymbolicCompression.Decompress()
 - ☐ AdaptiveHuffmanCompression.Compress()
 - ☐

- ☐ AdaptiveHuffmanCompression.Decompress()
- ☐ NeuralEmbeddingCompression.Compress()
- ☐ NeuralEmbeddingCompression.Decompress()
- ☐ ParallelNeuralSymbolicTraversal.Traverse()
- ☐ GPUAcceleratedTraversal.Traverse()
- ☐ MemoryOptimizedTraversal.Traverse()
- ☐ AdaptiveMemoryAwarePartitioning.Partition()
- ☐ NeuralBasedPartitioning.Partition()
- **Owner:** *Unassigned*

C3: Config Stub Methods

- **Status:** [] Not Started
- **File:** internal/config/config.go
- **Methods:**
 - ☐ UpdateConfigFromMap() (lines 566-568)
 - ☐ SaveTemplate() (lines 1464-1465)
 - ☐ LoadTemplate() (lines 1482-1483)
- **Owner:** *Unassigned*

C4: Hardcoded Model Alternatives

- **Status:** [] Not Started
- **File:** internal/llm/model_discovery.go:1136-1149
- **Issue:** Returns hardcoded list instead of querying APIs
- **Owner:** *Unassigned*

C5: Hardcoded Usage Analytics

- **Status:** [] Not Started
- **File:** internal/llm/usage_analytics.go:566,581,583
- **Issue:** Hardcoded placeholder metrics
- **Owner:** *Unassigned*

C6: Host Optimizer Stub

- **Status:** [] Not Started
- **File:** internal/cognee/host_optimizer.go:7-18
- **Issue:** Complete stub returning unchanged config
- **Owner:** *Unassigned*

C7: Tree-sitter Placeholder

- **Status:** [] Not Started
- **File:** internal/tools/mapping/treesitter.go:61-62,267-269
- **Issue:** Placeholder parser implementation
- **Owner:** *Unassigned*

C8: Silent Hardware Detection Failures

- **Status:** [] Not Started
- **File:** internal/hardware/detector.go:44-61
- **Issue:** Returns success despite all detection failures
- **Owner:** *Unassigned*

C9: Cache Error Hiding

- **Status:** [] Not Started
- **File:** internal/task/manager.go:257,263,302,307,337,343
- **Issue:** Returns nil,nil hiding cache errors
- **Owner:** *Unassigned*

C10: Transaction Verification Incomplete

- **Status:** [] Not Started
- **File:** internal/tools/multiedit/transaction.go:354-369
- **Issue:** Incomplete conflict verification
- **Owner:** *Unassigned*

C11: Zep Provider Query Format

- **Status:** [] Not Started
- **File:** internal/memory/providers/zep_provider.go:478
- **Issue:** Placeholder query format
- **Owner:** *Unassigned*

C12: Anima Provider Stub

- **Status:** [] Not Started

- **File:** internal/memory/providers/anima_provider.go:33-34
 - **Issue:** Stub AnimaClient
 - **Owner:** *Unassigned*
-

Test Coverage Tasks

Priority 1: Below 50% Coverage

Package	Current	Target	Status	Owner
internal/llm	44.9%	80%	[]	
internal/memory/providers	46.1%	80%	[]	
internal/workflow/planmode	39.7%	80%	[]	
shared/mobile-core	42.6%	80%	[]	
tests/e2e/challenges	49.0%	80%	[]	

Priority 2: 50-70% Coverage

Package	Current	Target	Status	Owner
internal/providers	51.7%	80%	[]	
internal/tools/mapping	53.8%	80%	[]	
internal/server	55.1%	80%	[]	
internal/tools	55.5%	80%	[]	
internal/tools/browser	55.1%	80%	[]	
internal/cognee	59.9%	80%	[]	
internal/tools/voice	60.2%	80%	[]	
internal/mcp	61.3%	80%	[]	
internal/focus	61.3%	80%	[]	
internal/tools/web	62.3%	80%	[]	
internal/editor/formats	62.6%	80%	[]	
internal/workflow	64.7%	80%	[]	
internal/workflow/autonomy	65.0%	80%	[]	
internal/tools/confirmation	65.0%	80%	[]	
internal/tools/filesystem	66.8%	80%	[]	

Package	Current	Target	Status	Owner
internal/rules	66.8%	80%	[]	
internal/hardware	67.5%	80%	[]	
internal/memory	67.8%	80%	[]	
internal/tools/shell	68.2%	80%	[]	
internal/llm/vision	68.3%	80%	[]	
internal/tools/multiedit	68.6%	80%	[]	
internal/agent/types	68.6%	80%	[]	
internal/redis	69.6%	80%	[]	

Priority 3: Applications (Very Low)

Application	Current	Target	Status	Owner
aurora-os	4.9%	50%	[]	
harmony-os	8.5%	50%	[]	
desktop	9.1%	50%	[]	
terminal-ui	9.6%	50%	[]	
cmd/cli	11.9%	50%	[]	

Priority 4: Untested LLM Providers

Provider	File	Status	Owner
Copilot	copilot_provider.go	[]	
KoboldAI	koboldai_provider.go	[]	
Llama.cpp	llamacpp_provider.go	[]	
Ollama	ollama_provider.go	[]	
OpenAI	openai_provider.go	[]	
OpenAI Compatible	openai_compatible_provider.go	[]	
OpenRouter	openrouter_provider.go	[]	
Tool Provider	tool_provider.go	[]	
xAI	xai_provider.go	[]	
Local Provider	local_provider.go	[]	

API Implementation Tasks

MCP Server Management Endpoints

Endpoint	Method	Status	Owner
/api/v1/mcp/servers	GET	[]	
/api/v1/mcp/servers	POST	[]	
/api/v1/mcp/servers/:id	PUT	[]	
/api/v1/mcp/servers/:id	DELETE	[]	
/api/v1/mcp/servers/:id/tools	GET	[]	
/api/v1/mcp/servers/:id/tools/:name/execute	POST	[]	

Workflow State/History Endpoints

Endpoint	Method	Status	Owner
/api/v1/workflows/:id/history	GET	[]	
/api/v1/workflows/:id/state	GET	[]	
/api/v1/workflows/:id/pause	POST	[]	
/api/v1/workflows/:id/resume	POST	[]	
/api/v1/workflows/:id/rollback	POST	[]	

Notification Management Endpoints

Endpoint	Method	Status	Owner
/api/v1/notifications	GET	[]	
/api/v1/notifications/rules	POST	[]	
/api/v1/notifications/channels	GET	[]	
/api/v1/notifications/test/:channel	POST	[]	
/api/v1/notifications/templates/:id	PUT	[]	

Feature Completion Tasks

Advanced Reasoning API

- **Status:** [] Not Started

- **Documentation:** CLI_Specs_5.md, Section 2, lines 71-82
- **Required:**
 - ☐ GenerateWithReasoning() interface
 - ☐ ReasoningRequest type
 - ☐ ReasoningResponse type
 - ☐ Chain-of-thought strategy
 - ☐ Tree-of-thoughts strategy
 - ☐ Provider implementations (Anthropic, OpenAI, Azure)
- **Owner:** *Unassigned*

CLI Commands

Command Group	Status	Subcommands	Owner
helix worker	[]	register, list, health, capabilities	
helix workflow	[]	history, pause, resume, rollback	
helix notify	[]	send, rules, channels, test	
helix session	[]	create, list, pause, resume, complete	

Configuration Additions

Config Section	Status	Owner
reasoning:	[]	
mcp.servers:	[]	
notifications: (centralized)	[]	

Documentation Tasks

Missing Package READMEs

Package	Status	Owner
internal/agent	[]	
internal/workflow	[]	

Package	Status	Owner
internal/session	[]	
internal/llm/vision	[]	
internal/project	[]	

Documentation Updates

Document	Task	Status	Owner
CLAUDE.md	Sync with implemented CLI commands	[]	
API_REFERENCE.md	Add missing endpoints	[]	
USER_MANUAL.md	Update configuration options	[]	
Video Courses	Create per VIDEO_COURSE_CURRICULUM.md	[]	
Website	Update per documentation	[]	

Progress Tracking

Daily Standup Format

Date: YYYY-MM-DD
Completed:
- [item]

In Progress:
- [item]

Blocked:
- [item] - reason

Next:
- [item]

Weekly Summary Format

Week: [number]
Critical Issues Fixed: X/12
Test Coverage Delta: +X.X%
API Endpoints Added: X/16

Tests Added: X

Documentation Pages Updated: X

Verification Commands

Run All Tests

```
cd HelixCode && go test ./...
```

Check Coverage

```
cd HelixCode && go test -cover ./...
```

Check Specific Package

```
cd HelixCode && go test -cover -v ./internal/llm/...
```

Generate Coverage Report

```
cd HelixCode && go test -coverprofile=coverage.out ./...
```

```
cd HelixCode && go tool cover -html=coverage.out -o coverage.html
```

Verify Build

```
cd HelixCode && make build
```

Run Linter

```
cd HelixCode && make lint
```

Notes

Session Resume Instructions

1. Read this file to understand current state
2. Check last “Daily Standup” entry
3. Continue from last “In Progress” items
4. Update tracker after completing items

Quality Verification

Each completed item must be verified: 1. Tests pass locally 2. Coverage improved (if applicable) 3. No new linter warnings 4. Documentation updated (if applicable) 5. Second review by team member

Tracker Version: 1.0 Last Updated: 2026-01-10